Code No: 114CN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, December - 2018
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART - A **(25 Marks)**

1.a)	What is the difference between direct and indirect address instruction?	[2]
b)	What is an interrupt?	[3]
c)	What is an I/O interface?	[2]
d)	Compare I/O bus Vs memory bus.	[3]
e)	What is cache coherence?	[2]
f)	Compare RAM with ROM.	[3]
g)	What is segment register of 8086?	[2]
h)	What are the roles of SP and BP registers of 8086?	[3]
i)	What are JNS and JNBE 8086 instructions?	[2]
j)	Write 8086 machine control instructions.	[3]

PART - B **(50 Marks)**

2. Explain various conditional branch instructions. [10]

OR

3. State and explain the various types of interrupts. [10]

4. Explain DMA mode of data transfer. [10]

OR

5. Explain the CPU-IOP communication. [10]

6.a) Explain the direct memory mapping of cache memory.
b) What is 'Write-through' and 'Write-back' of cache memory? [5+5]

OR

7. An address space is specified by 24 bits and the corresponding memory space by 16 bits.
a) How many words are there in the address space?
b) How many words are there in the memory space?
c) If a page consists of 2K words, how many pages and blocks are there in the system? [3+3+4]

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8. Draw and explain the pin diagram of 8086. [10]

OR

9. Explain the addressing modes of 8086. [10]

10. Write an 8086 assembly language program to find out the product of two matrices. [10]

OR

11. Write a 8086 assembly language program to find whether the given number 'n' is prime number or not. [10]

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